Md. Fardin Hossain

Computer Science and Engineering Student

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SUMMARY

Computer Science and Engineering student aspiring to become an AI Engineer, with experience in full-stack development, machine learning, and AI-powered applications. Skilled in Python and the MERN stack, with knowledge of deep learning, cybersecurity, and penetration testing. Passionate about building intelligent, scalable solutions to real-world problems.

EXPERIENCE

Machine Learning Intern | Internship — Jul 2026 - Oct 2026
FlyRank AI — *Remote · Dhaka, Bangladesh*

- Developed a Python-based content-refresh ranking system for **9,961 pages** using traffic decline and demand signals.
- Achieved **0.78 precision@50**, outperforming the fixed-rule baseline (**0.30**) on a client-grouped holdout.
- Processed a **78.8M-row** content-performance warehouse using DuckDB and delivered 12 executed notebooks plus a research paper.
- Completed the AI Fluency track, building an AI study-coach agent, automating workflows, and deploying a portfolio website with an impact-project capstone.

Technologies / Skills Used : *Python, scikit-learn, pandas, DuckDB*

PROJECTS

SchedOS — Interactive CPU Scheduling Laboratory | Github Website — Aug 2026 - Sep 2026
Lead Developer

- Built a client-side simulation lab with **React, TypeScript, and Tailwind CSS** that visualizes **8 CPU scheduling algorithms** (FCFS, SJF, SRTF, LJF, LRTF, Round Robin, Priority preemptive/non-preemptive) in real time.
- Engineered a zero-backend architecture with a tested simulation core, achieving **557 Vitest tests** passing.

Technologies / Tools Used : *React, TypeScript, Tailwind CSS, Vitest*

LocationKhuji - AI-Powered Location Discovery Platform | Website Github — Jun 2026 - Aug 2026
Lead Developer

- Built a full-stack **MERN platform** for discovering flats, hospitals, pharmacies, and services across **Bangladesh**.
- Implemented natural-language search in English, Bangla, and Banglish using **LLM APIs** (Groq, OpenRouter) with a multi-layer fallback architecture for reliability.
- Added interactive **Leaflet maps**, geospatial filtering, and role-based access (User, Owner, Admin) with owner/admin dashboards.

Technologies / Tools Used : *React, Node.js, MongoDB, Tailwind CSS, Express, Leaflet, Groq, OpenRouter*

SEU Routine Maker — University Routine Builder | Github Website — Jun 2026 - Jul 2026
Lead Developer

Built a **React/Next.js** tool that converts Southeast University UMS exports (HTML, PDF, screenshots) into a weekly class routine, with **OCR-based course-code detection**.

- Implemented timetable conflict detection, free-gap analysis, and browser class reminders.
- Enabled export to **.ics calendar, PDF, and PNG**.

Technologies / Tools Used : *React, Next.js, OCR*

UniBuddy - University Student Collaboration Platform | GitHub — Sep 2025 - Oct 2025
Lead Developer

- Built a cross-platform Flutter (Dart) app with an Appwrite backend: GPA/CGPA calculator with multi-semester tracking, cloud sync, and PDF grade-report export.
- Implemented authentication, user profiles, and offline support for Android, iOS, and Web.

Technologies / Tools Used : *Flutter, Dart, Appwrite*

SKILLS

Programming Languages :	Python, TypeScript, JavaScript, Dart, Java, C/C++
Machine Learning / AI :	Machine Learning, scikit-learn, pandas, PyTorch, Regression, Classification, Deep Learning, DuckDB, LLM APIs
Web & Mobile :	React, Next.js, Node.js, Express.js, Flutter, Tailwind CSS, REST APIs
Databases & Tools :	MongoDB, Firebase, Appwrite, Git, GitHub, Vercel, Linux
Cybersecurity :	Penetration Testing, Web Application Security, OWASP Top 10

EDUCATION

Southeast University | Tejgaon, Dhaka, Bangladesh — 2023 - 2027
Bachelor of Science (B.Sc.) in Computer Science and Engineering

CERTIFICATIONS

C|PTE: Certified Penetration Testing Engineer — 2026
Mile2 Cybersecurity Institute

IBM Generative AI Engineering Professional Certificate — 2026
IBM

AI Engineer for Developers Associate — 2026
DataCamp

Anthropic Academy — 20 certificate courses completed — 2026
Anthropic

LANGUAGES

Bengali (Native), English (Professional)